
(a)
$|0\rangle$
$|\uparrow\downarrow\rangle$
$|\uparrow\rangle$
$|\downarrow\rangle$
$\sum_\sigma f^\dagger_{i\sigma} f_{i\sigma} = 1$
(b)
$J\,\boldsymbol{S}_i \cdot \boldsymbol{S}_j$
$\chi^\dagger_{ij}\chi_{ij}$
the same bond, in fermions
(c)
$\varepsilon(k)$
$-k_F$
k_F
k
spinon band, filled up to $\pm k_F = \pm\pi/2$